

Name: _____

Class: _____

ACTIVITY SHEET

1.4 Identifying different types of elements

The known elements can be classified as: halogens, noble gases, other non-metals, metalloids, alkali metals, alkaline earth metals, transition metals, post-transition metals, lanthanoids and actinoids.

Instructions

- 1 Research the meaning of any terms from the list above with which you are unfamiliar.
- 2 Find out which elements belong to each class.
- 3 Colour code the periodic table below to show the location of each class of elements.
- 4 Use a key to indicate which class each colour represents.

hydrogen 1 H 1.008												helium 2 He 4.003																							
lithium 3 Li 6.941		beryllium 4 Be 9.012												boron 5 B 10.811		carbon 6 C 12.011		nitrogen 7 N 14.007		oxygen 8 O 15.999		fluorine 9 F 18.998		neon 10 Ne 20.180											
sodium 11 Na 22.990		magnesium 12 Mg 24.305												aluminium 13 Al 26.982		silicon 14 Si 28.086		phosphorus 15 P 30.974		sulfur 16 S 32.066		chlorine 17 Cl 35.453		argon 18 Ar 39.948											
potassium 19 K 39.098		calcium 20 Ca 40.078		scandium 21 Sc 44.956		titanium 22 Ti 47.88		vanadium 23 V 50.942		chromium 24 Cr 51.996		manganese 25 Mn 54.938		iron 26 Fe 55.933		cobalt 27 Co 58.933		nickel 28 Ni 58.693		copper 29 Cu 63.546		zinc 30 Zn 65.39		gallium 31 Ga 69.732		germanium 32 Ge 72.61		arsenic 33 As 74.922		selenium 34 Se 78.09		bromine 35 Br 79.904		krypton 36 Kr 84.80	
rubidium 37 Rb 84.468		strontium 38 Sr 87.62		yttrium 39 Y 88.906		zirconium 40 Zr 91.224		niobium 41 Nb 92.906		molybdenum 42 Mo 95.94		technetium 43 Tc 98.907		ruthenium 44 Ru 101.07		rhodium 45 Rh 102.906		palladium 46 Pd 106.42		silver 47 Ag 107.868		cadmium 48 Cd 112.411		indium 49 In 114.818		tin 50 Sn 118.71		antimony 51 Sb 121.760		tellurium 52 Te 127.6		iodine 53 I 126.904		xenon 54 Xe 131.29	
cesium 55 Cs 132.905		barium 56 Ba 137.327		57-71		hafnium 72 Hf 178.49		tantalum 73 Ta 180.948		tungsten 74 W 183.85		rhenium 75 Re 186.207		osmium 76 Os 190.23		iridium 77 Ir 192.22		platinum 78 Pt 195.08		gold 79 Au 196.967		mercury 80 Hg 200.59		thallium 81 Tl 204.383		lead 82 Pb 207.2		bismuth 83 Bi 208.980		polonium 84 Po 209		astatine 85 At 210		radon 86 Rn 222	
francium 87 Fr 223		radium 88 Ra 226		88-103		rutherfordium 104 Rf [261]		dubnium 105 Db [262]		seaborgium 106 Sg [266]		bohrium 107 Bh [264]		hassium 108 Hs [269]		meitnerium 109 Mt [268]		darmstadtium 110 Ds [269]		roentgenium 111 Rg [272]		copernicium 112 Cn [277]				flerovium 114 Fl [289]				livermorium 116 Lv [293]					
lanthanum 57 La 138.906		cerium 58 Ce 140.115		praseodymium 59 Pr 140.908		neodymium 60 Nd 144.24		promethium 61 Pm 144.913		samarium 62 Sm 150.36		europium 63 Eu 151.966		gadolinium 64 Gd 157.25		terbium 65 Tb 158.925		dysprosium 66 Dy 162.50		holmium 67 Ho 164.930		erbium 68 Er 167.26		thulium 69 Tm 168.934		ytterbium 70 Yb 173.04		lutetium 71 Lu 174.967							
actinium 89 Ac 227		thorium 90 Th 232		protactinium 91 Pa 231		uranium 92 U 238		neptunium 93 Np 237		plutonium 94 Pu 244		americium 95 Am 243		curium 96 Cm 247		berkelium 97 Bk 247		californium 98 Cf 251		einsteinium 99 Es [254]		fermium 100 Fm 257		mendelevium 101 Md 258		nobelium 102 No 259		lawrencium 103 Lr [262]							

Key:

- 5 Label groups 1 to 18 across the top of the periodic table.
- 6 Label periods 1 to 7 down the left side of the periodic table.